

- This exam contains 5 pages (including the front and back of this cover page) and 10 problems. Check to see if any pages are missing.
- This is the first exam and will count for 20% of your final grade. You have until Monday, March 21st at 11:59 PM to complete this exam. You may use consult your notes and textbook during this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 140 points total in this exam, as well as 1 bonus questions worth a total of 10 possible points.

Page:	2	3	4	5	Total
Points:	18	22	80	20	140
Score:					

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

(a) ☐ 2 points If $z = x + iy$, then $\sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$ and $\cos z = \frac{1}{2}(e^{iz} + e^{-iz})$.

☐ True☐ False

(b) ☐ 2 points The function $f : \mathbb{C} \rightarrow \mathbb{C}$ given by $f(z) = \cos(z)$ is a bounded function (i.e. there exists a value $M \geq 0$ such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$.)

☐ True☐ False

(c) ☐ 2 points If $z = r(\cos \theta + i \sin \theta)$ and n is a positive integer, then $z^n = r^n(\cos n\theta + i \sin n\theta)$

☐ True☐ False

(d) ☐ 2 points If $z = x + iy$, then $e^z = e^x(\cos y + i \sin y)$.

☐ True☐ False

(e) ☐ 2 points If $U \subseteq \mathbb{C}$ is open, and $f : U \rightarrow \mathbb{C}$ has continuous partial derivatives at some point $z_0 \in U$, then f is analytic at z_0 .

☐ True☐ False

(f) ☐ 2 points The function $f : \mathbb{C} \rightarrow \mathbb{C}$ given by $f(z) = e^z$ is injective (i.e. one-to-one).

☐ True☐ False

(g) ☐ 2 points Given topological spaces X and Y , a function $f : X \rightarrow Y$ is continuous if and only if for all closed sets $K \subseteq Y$, the preimage $f^{-1}(K) \subseteq X$ is closed.

☐ True☐ False

(h) ☐ 2 points $\operatorname{Im} \left(\int_{\gamma} f dz \right) = \int_{\gamma} (\operatorname{Im} f) dz$.

☐ True☐ False

(i) ☐ 2 points $|a^b| = |a|^{|b|}$ for all $a, b \in \mathbb{C}$.

☐ True☐ False

2. 10 points Find $x, y \in \mathbb{R}$ such that

$$\frac{9-i}{i+7} + \frac{3}{(1+i)^2} = x + iy.$$

3. Let z be a nonzero complex number.

- (a) 2 points What is the definition of z^i ?
- (b) 5 points Show that if one value of z^i is real, then all other values are real.
- (c) 5 points If z^i is real for some (all) values, what are the possible values of z ?

4. 20 points A region A is called *star-shaped* if there exists a point $z_0 \in A$ such that for any $z \in A$, the line segment between z_0 and z lies in A . Prove every star-shaped region is simply connected.
5. 20 points Let γ be a curve in whose image lies in a simply connected region A and suppose z_0 is a point not in A . Prove that $I(\gamma; z_0) = 0$.
6. 10 points Let γ be the straight line segment joining 1 and $2 + i$. Evaluate

$$\int_{\gamma} z^2 + 2z + 3dz.$$

7. 20 points Let $u(x, y) = y^3 - 3x^2y$. Show that u is harmonic. Then, determine a conjugate function $v(x, y)$ such that $u + iv$ is analytic.

8. 10 points Let γ be the trapezoid with corners $3 \pm 3i$, $-2 \pm 2i$ traversed counterclockwise. Evaluate

$$\int_{\gamma} \frac{\sin e^{2z+1}}{z^3 - 2z^2 + 4z - 8}.$$

(Hint: Homework 7, Problem 1.a)

9. 20 points Let γ be the circle of radius 2 centered at the origin, traversed counterclockwise. Evaluate

$$\int_{\gamma} \left(z + \frac{1}{z} \right)^2.$$

10. 10 points (bonus) The open unit disk $\mathbb{D}$ is the set of complex numbers z with $|z| < 1$. The upper half plane H is the set of complex numbers z with $\text{Im}(z) > 0$. Find an analytic map $f : \mathbb{D} \rightarrow H$ which is one-to-one and onto.